

Datum Plot Guide

Analyse your data and make publication-ready graphs, step by step.

Datum Plot 0.4.0 · 2026-11-10
github.com/Edgar-Mendonca/Datum-Plot

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Welcome

Datum Plot is a program for Windows that analyses scientific data and makes publication-ready graphs. You type, paste or import your data into a table; choose an analysis; and Datum Plot shows the results with a plain-language answer, draws the graph and lays out the figure for a journal, a poster or a slide. Everything stays linked: change a value in the table and every result and graph that uses it updates.

The calculations and the drawing are done by Python (numpy, SciPy and matplotlib) inside the app. Nothing else needs to be installed, and your data never leaves your computer.

This guide takes you through Datum Plot step by step, from installing it to an exported figure. You do not need to read it in one go: each chapter stands on its own.

How to read this guide:

- Names in **bold** are tabs, buttons and boxes exactly as they appear in Datum Plot.
- An arrow › is a path: **Analyze › Test Chooser** means the **Analyze** tab, then the **Test Chooser** button.
- Keys are shown like this: `Ctrl+S`.
- The orange numbers on the pictures match the numbered lists under them.

> **Tip.** Almost every tool in Datum Plot has a small **?** beside its name. Click it for a short card that says what the tool does and how to use it.

Install, update and start

Install

1. Download `DatumPlot-x.y.z-setup.exe` from the Releases page: github.com/Edgar-Mendonca/Datum-Plot/releases.
2. Run it. It installs for your own Windows account, so you do not need administrator rights.
3. The first time, Windows may show **Windows protected your PC**, because the installer is not code-signed yet. Click **More info**, then **Run anyway**.

The first start takes a little longer while Datum Plot prepares its Python engine. The status bar says **Ready** when the engine has started.

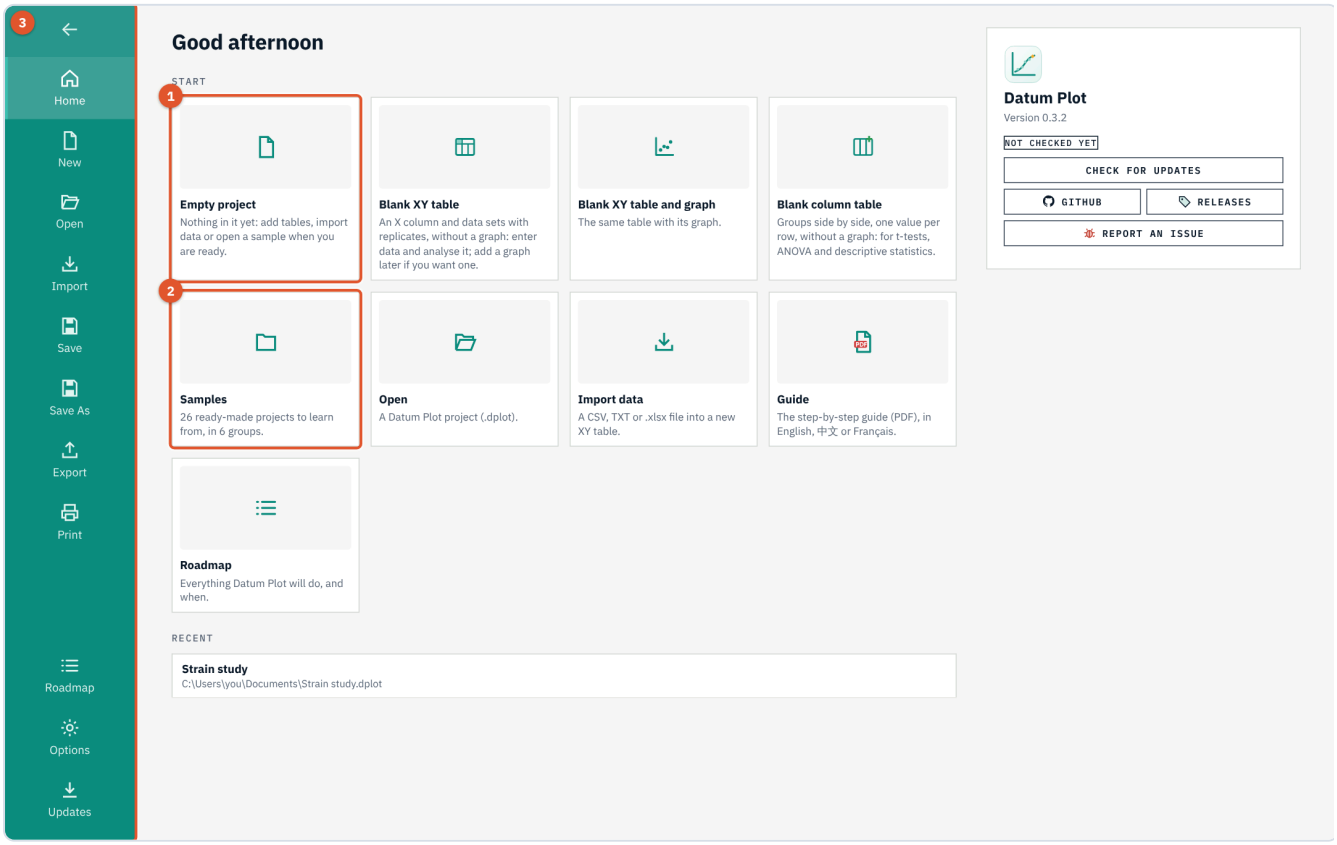
Updates

Datum Plot checks for a new version when it starts. When one is ready, an **Update x.y.z available** button appears in the title bar; click it, or open **File › Updates** and click **Update now**. The download is checked against its checksum before it installs, and Datum Plot restarts by itself. Your projects are not touched.



File › Updates: the version, the update check and the links to GitHub.

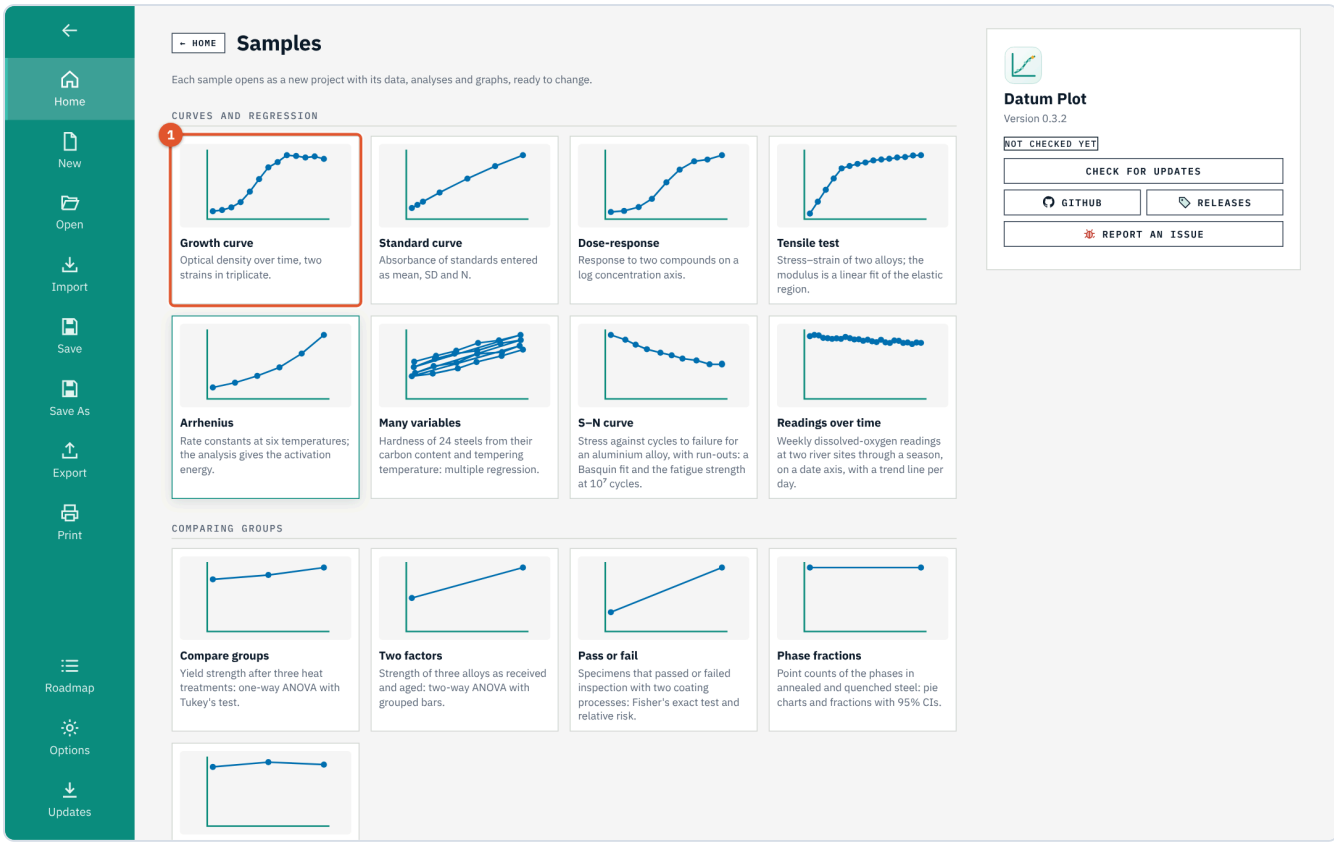
The Home page



The Home page (File › Home).

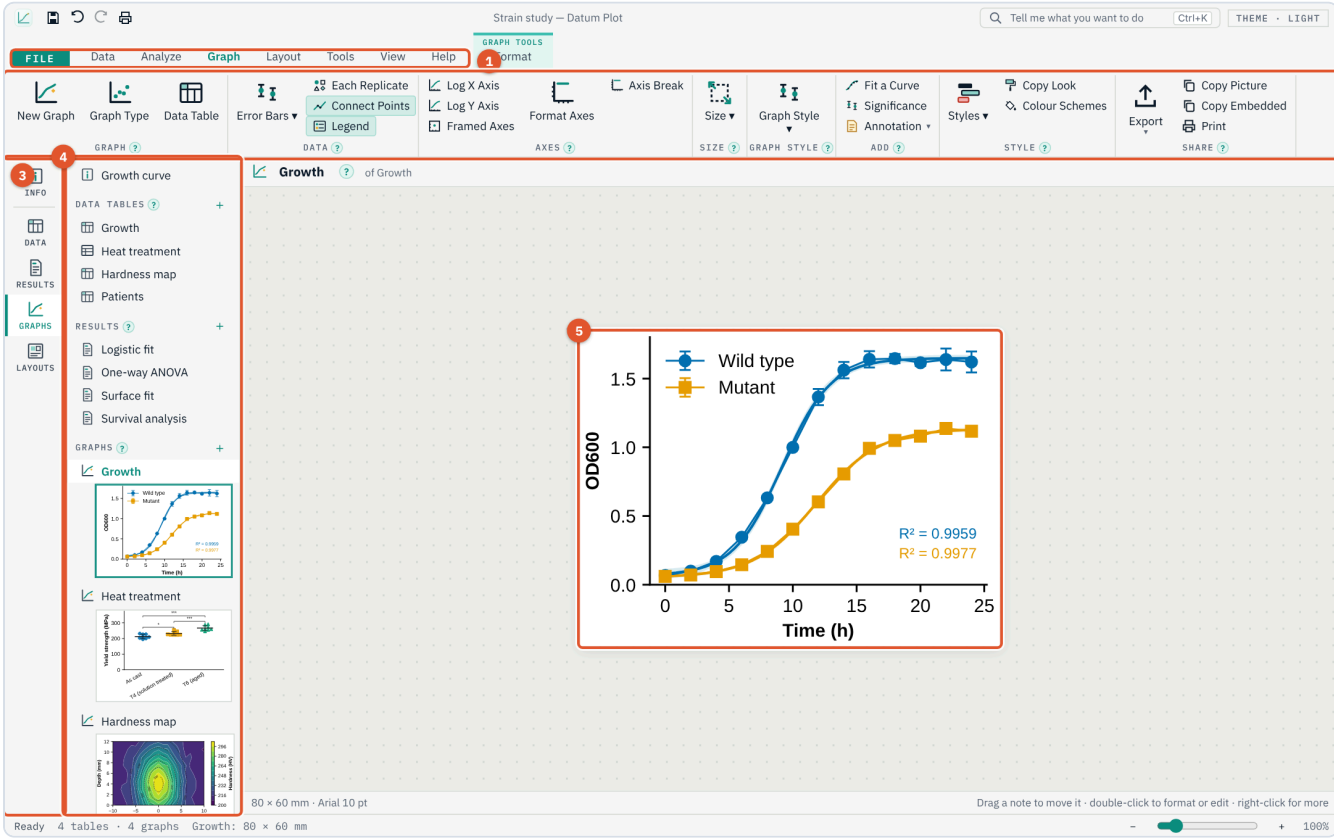
1. **Starts.** Datum Plot opens here. **Empty project**: nothing in it yet. **Blank XY table**: a table without a graph, to enter and analyse data only. **Blank XY table and graph**: the same with its graph. **Blank column table**: groups side by side, without a graph, for t-tests and ANOVA. A graph can be added later with **Graph › New Graph**.
2. **Samples.** A folder of small, complete projects with real-looking data, an analysis and a graph. It opens inside Home, with the samples under headings: curves and regression, comparing groups, survival and reliability, medical tests, signals and thermal analysis, maps and special graphs. Open one to see how a finished project looks, then replace the data with yours; **← Home** goes back.
3. **The File page. New**, **Open**, **Import**, **Save**, **Save As**, **Export** and **Print**, and below them **Roadmap**, **Options** and **Updates**. The arrow at the top goes back to the project.

Beside them are **Open**, **Import data**, the **Guide** (this guide, in English, Chinese or French) and the **Roadmap**; further down, your recent projects.



Samples, under their headings.

The window at a glance



The Datum Plot window with a graph open.

1. **Tabs. File** opens the File page. **Data**, **Analyze**, **Graph**, **Layout**, **Tools**, **View** and **Help** each show their own ribbon.
2. **Ribbon.** The tools of the current tab, in named groups. When the window is narrow (a laptop screen), groups fold from the right into one button each with the group's name, such as **Style ▾**: click it for its commands. On short screens the File page lists its commands in a compact column.
3. **Sidebar.** Jumps between the project's **Info** sheet, **Data**, **Results**, **Graphs** and **Layouts**.
4. **Navigator.** Every data table, result, graph and layout in the project, with small pictures of the graphs. Right-click an item to rename, duplicate or delete it.
5. **The open item.** Here a graph; it can also be a data table, a results sheet or a page layout.

The box **Tell me what you want to do** at the top (or `Ctrl+K`) finds any command by name: type "log axis" or "t-test" and press `Enter`.

How a project is linked

A project holds data tables, analyses, graphs and layouts:

- A **data table** holds the numbers.
- An **analysis** reads a data table and gives a **results sheet**.
- A **graph** draws a data table, and can show the curves and significance stars of its analyses.

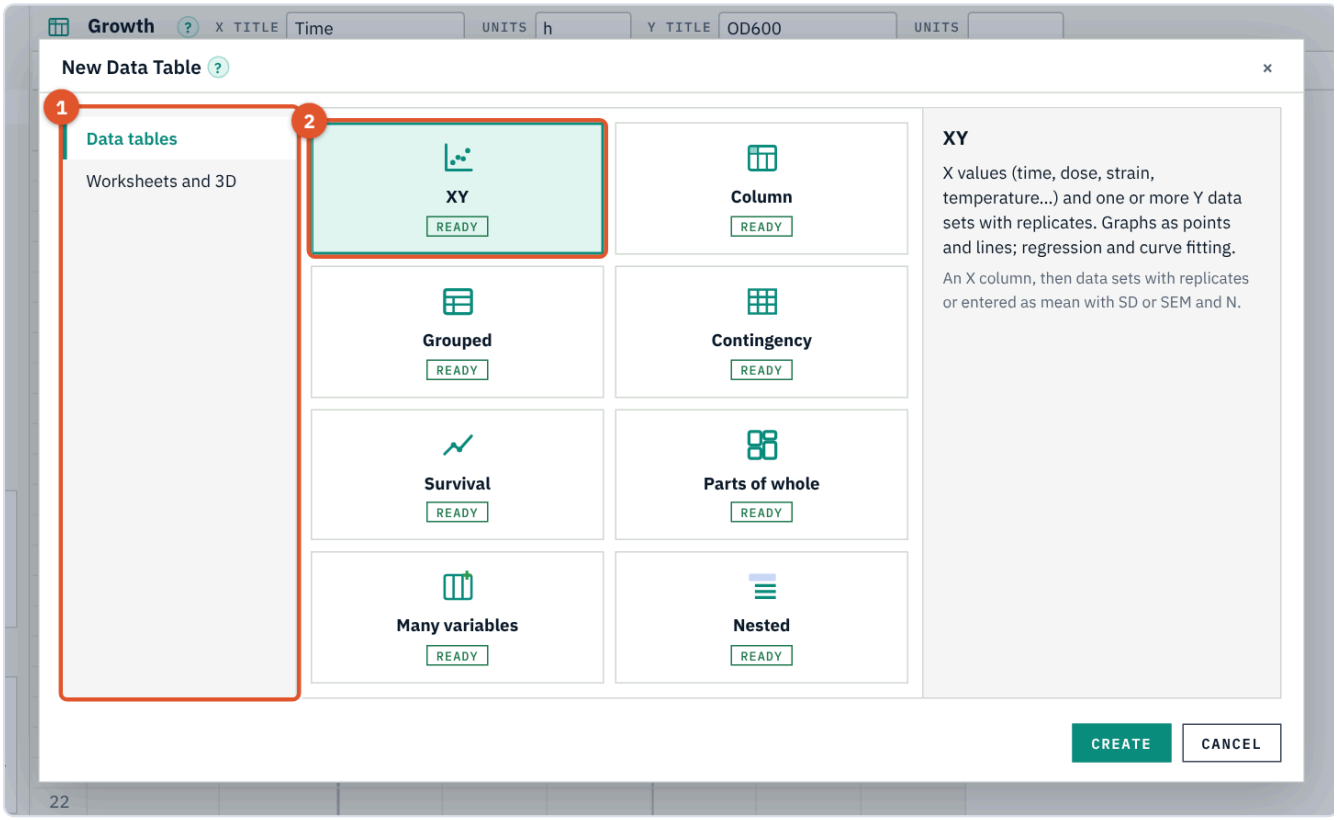
- A **layout** places graphs and text on a page.

When you change a value, every result and graph that depends on it is worked out again. You never need to "refresh".

Data tables

Kinds of table

Choose the kind of table that matches your experiment: **Data › New Table**.



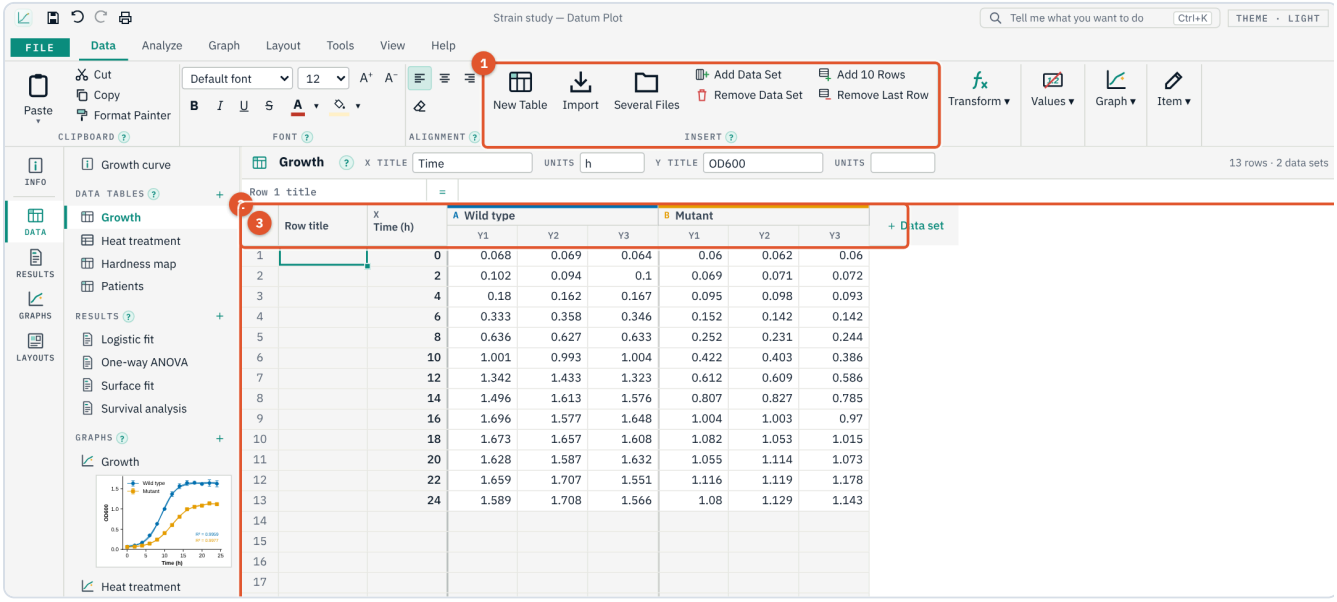
Data › New Table: choose a kind of table.

1. **Groups of tables.** Data tables, and Worksheets and 3D.
2. **The kinds.** Click one to read what it is for; double-click to create it.

Kind	What it holds	Typical analyses
XY	X values (time, dose, strain...) and Y data sets with replicates	regression, curve fitting
Column	each column is a group	t-tests, ANOVA, Bland-Altman, ROC
Grouped	two factors: rows and data sets	two-way ANOVA
Contingency	counts of outcomes per group	chi-square, Fisher's exact test
Survival	one row per subject: time, and 1 (event) or 0 (censored)	Kaplan-Meier, log-rank
Parts of whole	parts of a total	fractions, pie and donut
Many variables	one row per sample, many properties	multiple regression, PCA, clustering

Kind	What it holds	Typical analyses
Nested	groups with subgroups (batches, animals)	nested t-test and ANOVA
Worksheet	free columns with roles, units and formulas	raw instrument data
Matrix	Z values on an X–Y grid	contour maps, 3D surfaces, surface fits

Entering data



A data table (Data tab).

1. **Data › Insert. New Table**, **Import**, **Several Files**, **Add Data Set**, **Remove Data Set** and rows.
2. **Headings.** The X title and units, and each data set's title. Double-click a heading to change it, the replicates or the way values are entered. Long titles wrap onto up to three lines. Drag the right edge of a heading to widen its column; double-click the edge to fit the title and values.
3. **Cells.** Type a value and press `Enter`. Copy and paste work with spreadsheets (`Ctrl+C`, `Ctrl+V`); `Ctrl+D` fills down.

A data set can be entered as replicates (each value in its own column) or as **mean, SD and N** or **mean, SEM and N** from a paper or another program: choose it in the data set's heading.

To leave a value out of graphs and analyses without deleting it, select it and press `Ctrl+E` (**Exclude / Include**). Excluded values are shown in blue italics, struck through.

Formulas and transforms

- **Data › Column Formula** fills a column from the others, for example `(B - A) / A * 100`. The column updates when the data change. Dates work too: `date(2026, 11, 5)`, `today()`, `year(X)`, `month(X)`, `day(X)`, and `days(B, A)` or `hours(B, A)` for the time between two dates.

- **Data › Transform** makes a new table of transformed values: logarithms, 1/X, °C to K, a baseline subtracted, normalised.
- **Data › Check Data** looks for typing mistakes, missing values and values a log axis cannot show.

Import

Data › Import (or **File › Import**) reads CSV, TXT and TSV files, .xlsx workbooks and .pzfx projects. A preview shows how the columns will be read; choose the header row and the replicates and click **Import**. When the X column holds dates, **The X column holds dates** is ticked for you. **Data › Several Files** imports many runs at once: one data set per file in one table when they share their X values, otherwise one table per file. Plate-reader layouts (96 and 384 wells) are recognised.

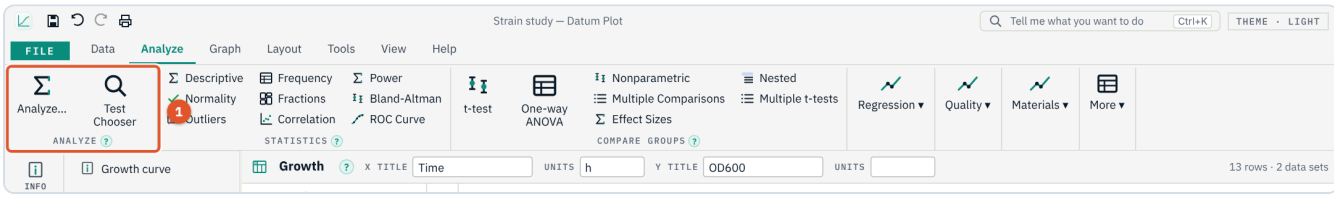
Dates and times

The X column of an XY table can hold dates. Right-click its heading (or choose **Format X Column** from the right-click menu of a cell) and pick **Dates** or **Dates and times**, and how they are shown: `5 Nov 2026`, `05-Nov-26`, `2026-11-05`, `05/11/2026` or `Nov 2026`.

- Pasting dates into a number X column asks whether to treat the column as dates.
- Type or paste dates as `2026-11-05`, `5-Nov-26`, `5 Nov 2026`, `Nov 5, 2026` or `05/11/2026`, with a time if you like (`2026-11-05 14:30`). **File › Options › Dates** says whether `05/11/2026` is day first or month first.
- Dates are kept as numbers, days since 1 January 1970, so graphs and fits work on them: the graph gets a date axis (**Format Graph › X Axis › Tick labels**: automatic, day and month, month and year, ISO or week number; **Major ticks every** day, week, 2 weeks, month, 3 months or year), and a straight-line fit gives its slope **per day**, the X-intercept as a date and the value at the first date.
- **Format Graph › Notes › Line at a date** draws a dashed line across the graph at a date you type, with a label.
- Workbook exports (.xlsx, .ods) write real dates. Files from Datum Plan open with their dates already set.

Analyses and results

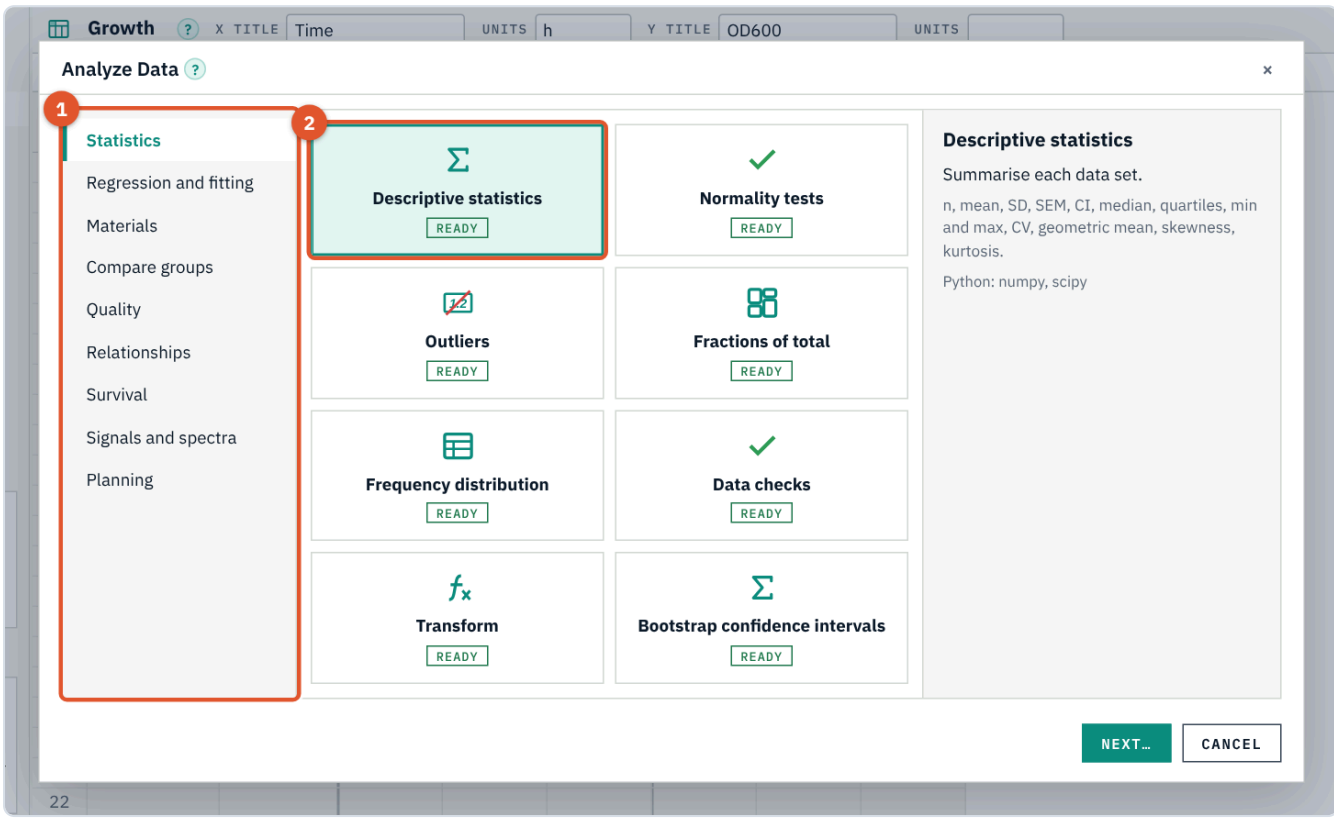
Choosing an analysis



The Analyze tab.

1. **Analyze... and Test Chooser. Analyze...** lists every analysis; **Test Chooser** asks three questions about your data and suggests the right test.

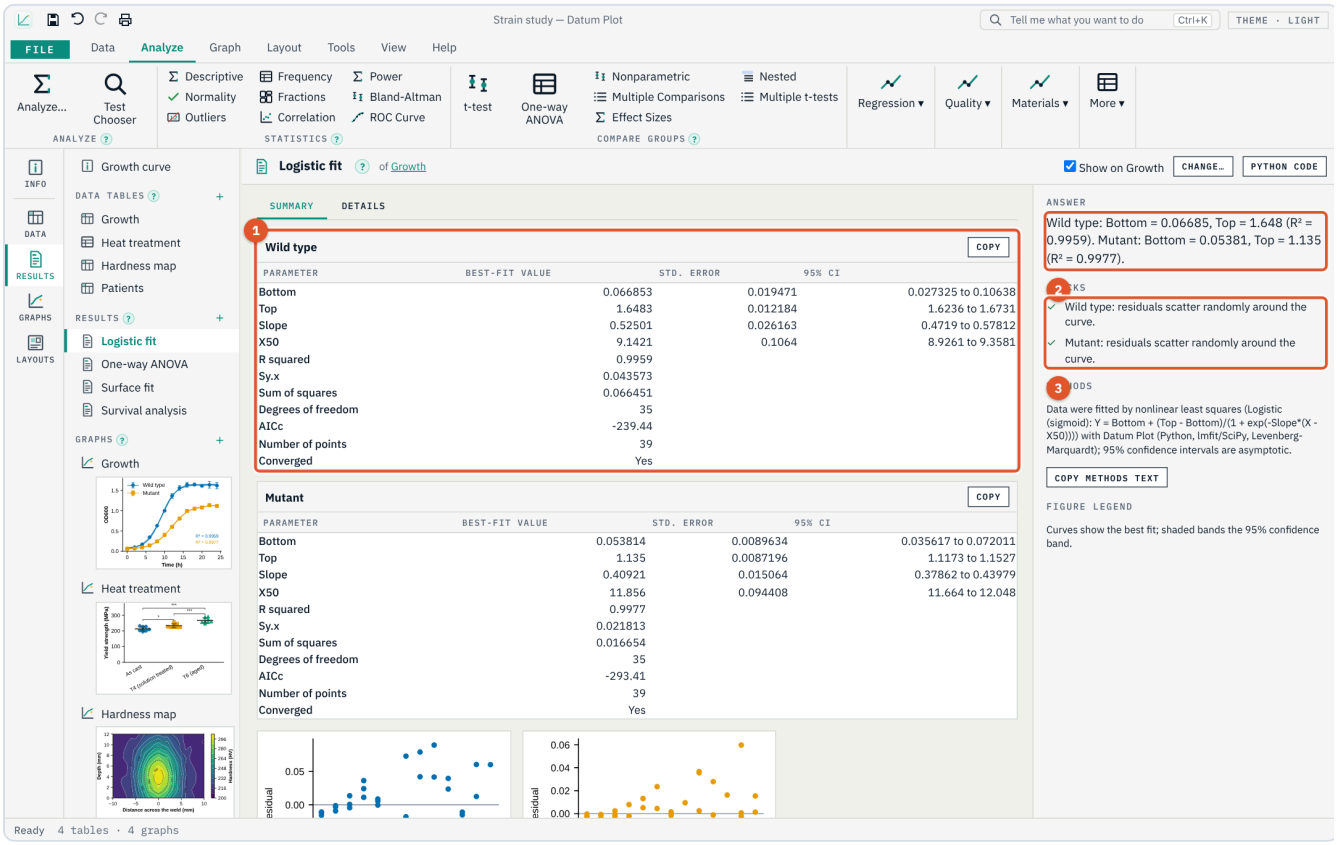
The other groups of the Analyze tab start an analysis directly: **Statistics**, **Compare Groups**, **Regression**, **Materials** and more.



Analyze...: every analysis, by group.

1. **Groups.** Statistics, regression and fitting, materials, compare groups, relationships, survival, signals and spectra.
2. **Analyses.** Click one to read what it does and which Python libraries it uses; **Next...** asks which data sets to include and the options.

The results sheet



A results sheet: a logistic curve fit.

1. **Result tables.** The numbers, with standard errors and 95% confidence intervals. **Copy** copies a table for a spreadsheet or a document.
2. **The answer.** What the analysis found, in plain words.
3. **Checks.** Whether the assumptions hold (for example, whether residuals scatter randomly), with a tick or a warning.

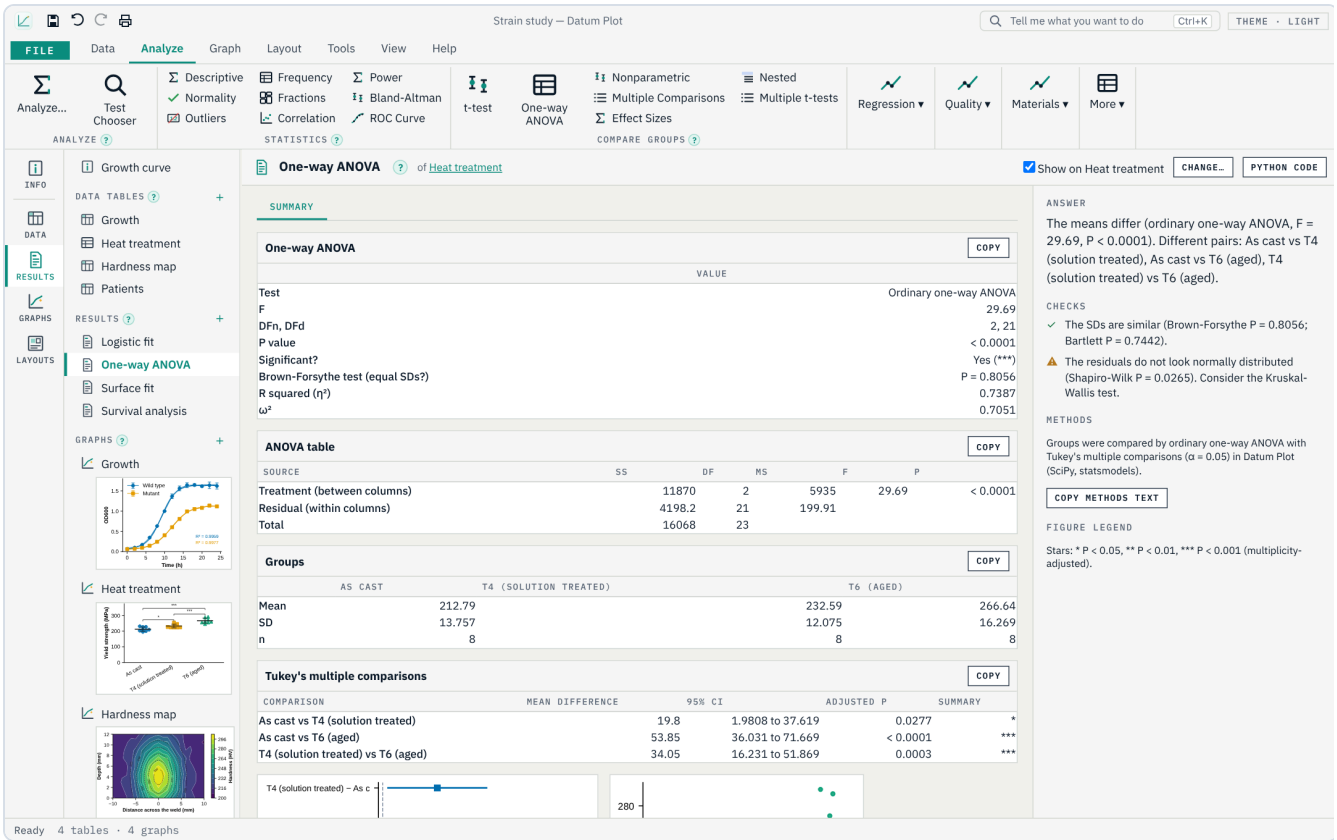
Below them are the **Methods** text, ready to paste into a paper (**Copy methods text**), and the figure legend. **Change...** reopens the options; **Python code** shows the Python behind the result so you can repeat it elsewhere. Tick **Show on graph** to draw a fit or the significance stars on the graph.

> **Tip.** Not sure what a number means? Click **?** beside it, or open **Help › Glossary**.

What Datum Plot can analyse

- **Statistics:** descriptive statistics, normality tests, outliers, frequency distributions, fractions of a total, bootstrap confidence intervals of the mean, median or SD.
- **Compare groups:** t-tests (unpaired, Welch, paired), nonparametric tests, one-way and two-way ANOVA with multiple comparisons, repeated measures and mixed-effects models, multiple t-tests with the false discovery rate and volcano plots, nested t-tests and ANOVA, effect sizes, power, three-way ANOVA, ANCOVA (groups compared allowing for a covariate).

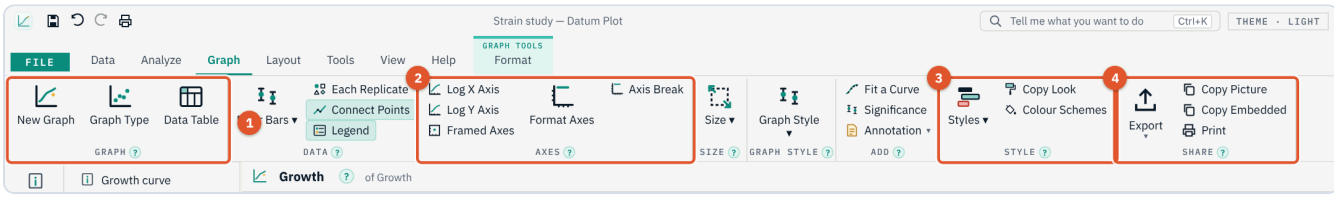
- **Relationships:** correlation, Bland-Altman method comparison, ROC curves, contingency tables (chi-square, Fisher, relative risk, odds ratio), agreement of paired ratings (Cohen's kappa, McNemar's and Bowker's tests).
- **Regression and fitting:** linear regression, curve fitting from a model library or your own equation, global fits and model comparison, interpolation, area under the curve, smoothing, multiple, logistic and Poisson regression, principal components and clustering, surface fits, smoothing splines.
- **Survival:** Kaplan-Meier curves, the log-rank test and hazard ratios.
- **Signals and spectra:** peak finding, Gaussian, Lorentzian and Voigt peak fits, Fourier spectra and filters, and how the frequencies change over time: a **wavelet scalogram** or a **spectrogram** (Signals › Operation), a heat map of time against frequency with the strongest frequency at each time as a table to graph.
- **Quality:** control charts (individuals, X̄–R, X̄–S, p) with rule checks and Cp/Cpk; **Analyze › New Design** makes a two-level factorial design (full or half fraction) to fill in, and Factorial Analysis gives the effects, a Pareto chart and the main effects.
- **Materials:** tensile tests, Weibull strength, Arrhenius activation energies, S–N fatigue curves, DSC and TGA thermal analysis.



A one-way ANOVA results sheet with multiple comparisons.

Graphs

The Graph tab



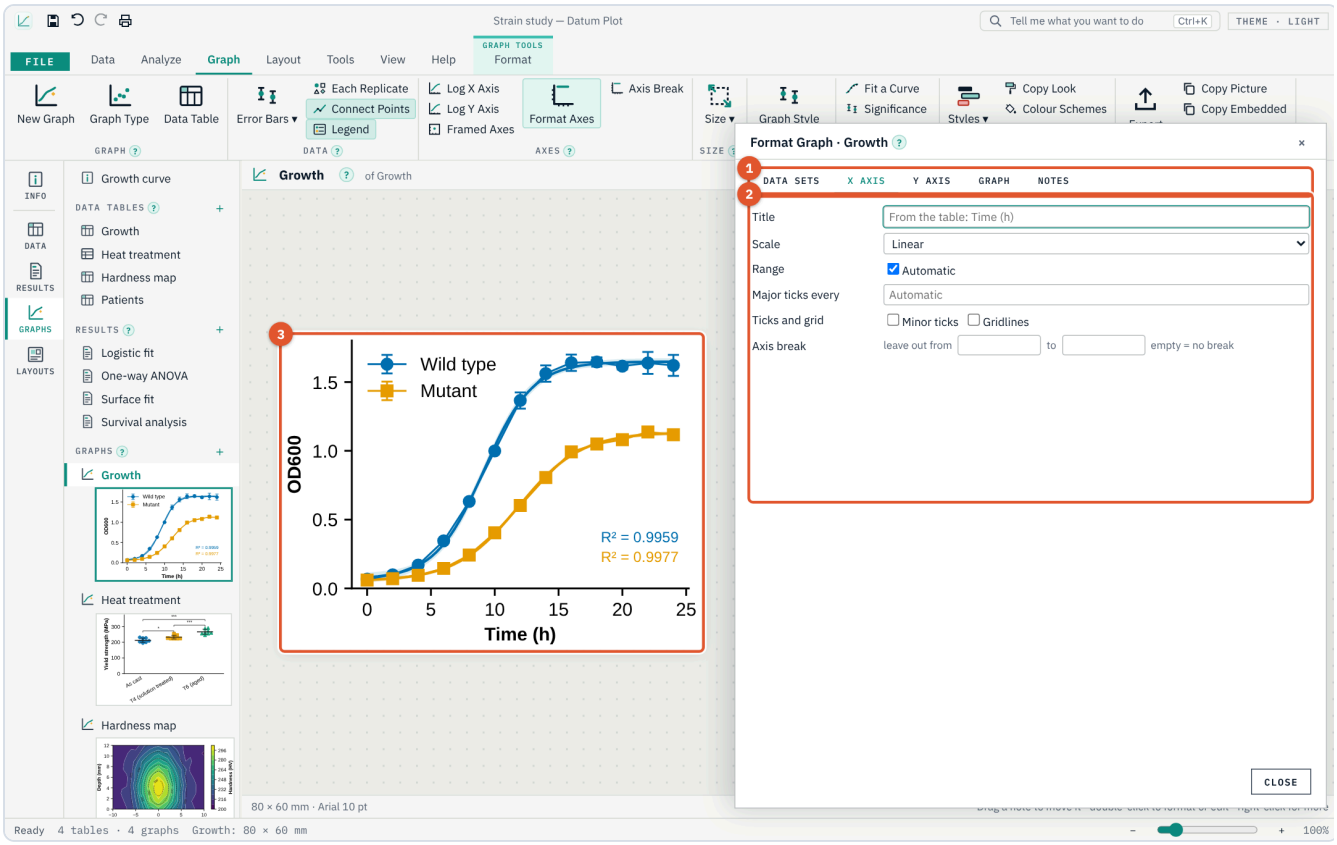
The Graph tab.

1. **Graph. New Graph**, **Graph Type** (scatter, bars, box, violin, before–after, lines and more) and **Data Table** (back to the data).
2. **Axes.** Logarithmic axes, framed axes, **Format Axes** and **Axis Break**.
3. **Style.** Ready-made **Styles** for journals and presentations, **Copy Look** to other graphs, and **Colour Schemes** (including colour-blind-safe ones).
4. **Share. Export**, **Copy Picture** (paste straight into a document or slides), **Copy Embedded** (a graph the document can open again in Datum Plot) and **Print**.

The other groups set the **Error Bars** (the mean with SD, SEM, 95% CI or range, or the median with its interquartile range, range or 95% CI), show **Each Replicate**, connect points, show the legend (on a column graph it names each group by its colour), set the **Size** and add a curve fit, significance stars or annotations. With an **Axis Break**, **Format Graph › Y Axis** also sets how much of the axis each part takes and the ticks beyond the break.

Formatting a graph

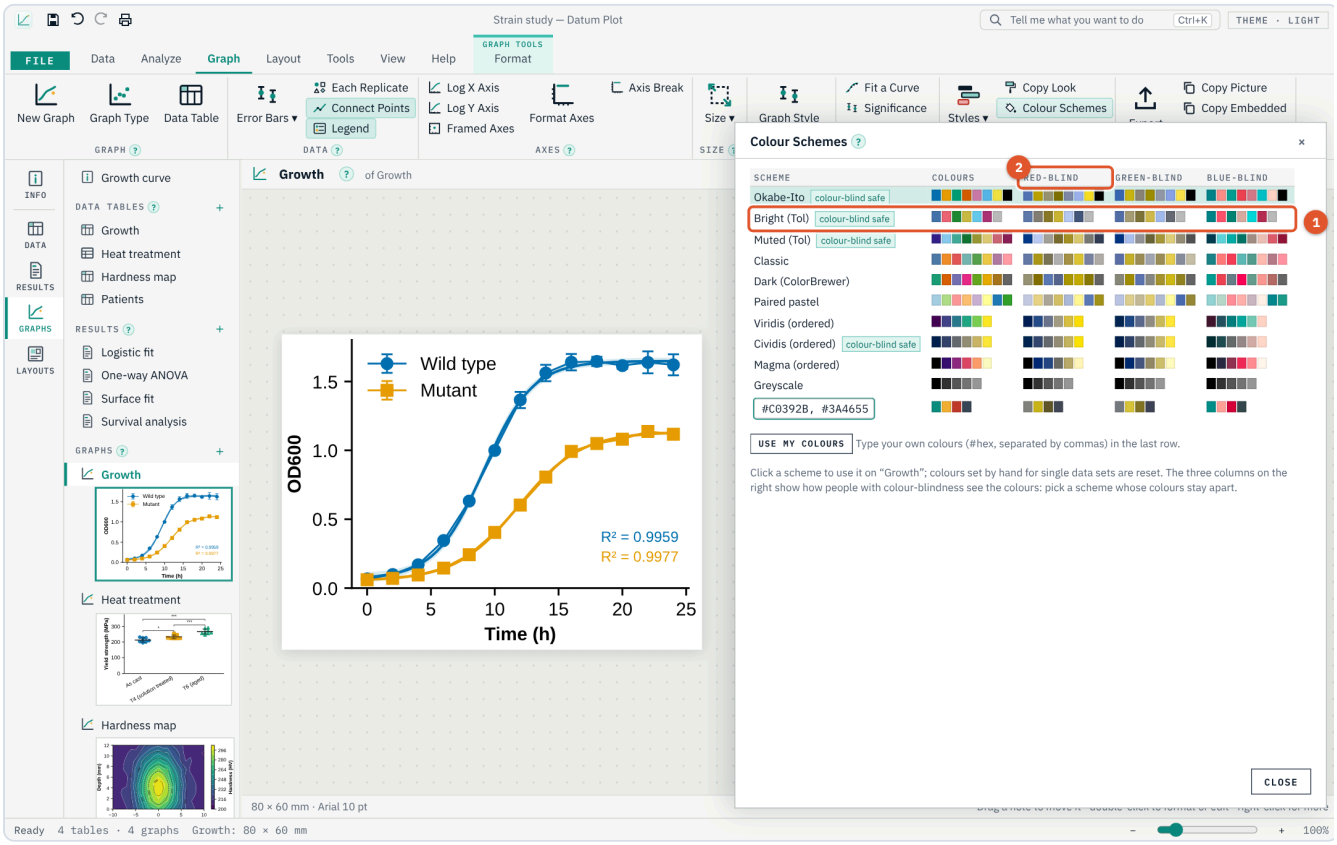
Double-click any part of a graph (an axis, a data set, the title) to format it, or right-click an axis › **Format Axes**, or use **Graph › Format Axes**. There is one Format Graph window: these commands switch it to the page you asked for, and it always works on the graph you are looking at (open another graph and it follows; open a table or results and it closes). The graph's name is shown above it; in the navigator each graph's preview sits under its name (right-click › **Hide Previews** to save space). Drag the navigator's right edge to widen it. Ctrl+click (or Shift+click) selects several items: right-click them to delete them together, move them to a folder (**Move to Folder › New Folder...**) or colour their names (**Colour**). Right-click a data table › **Show Only This Family** to see just that table with its results, graphs and layouts; **Show All** brings back the rest. The + beside Graphs makes a new graph of any table or copies the open graph with its formatting; right-click a table › **Duplicate with Graphs and Results** or **Duplicate Table Only**; right-click a layout › **Copy Picture** or **Export**. In **Format Graph › Data Sets**, **Order** moves data sets along a column graph (the table keeps its order), and **Fill** and **Border** colour bars, boxes and violins.



Format Graph, docked beside the graph.

1. **Pages. Data Sets** (symbols, colours, lines, error bars, the right Y axis), **X Axis**, **Y Axis**, **Graph** (size and fonts) and notes.
2. **Settings.** Changes apply at once.
3. **The graph** stays visible while you work.

Colours



Graph › Colour Schemes, beside the graph.

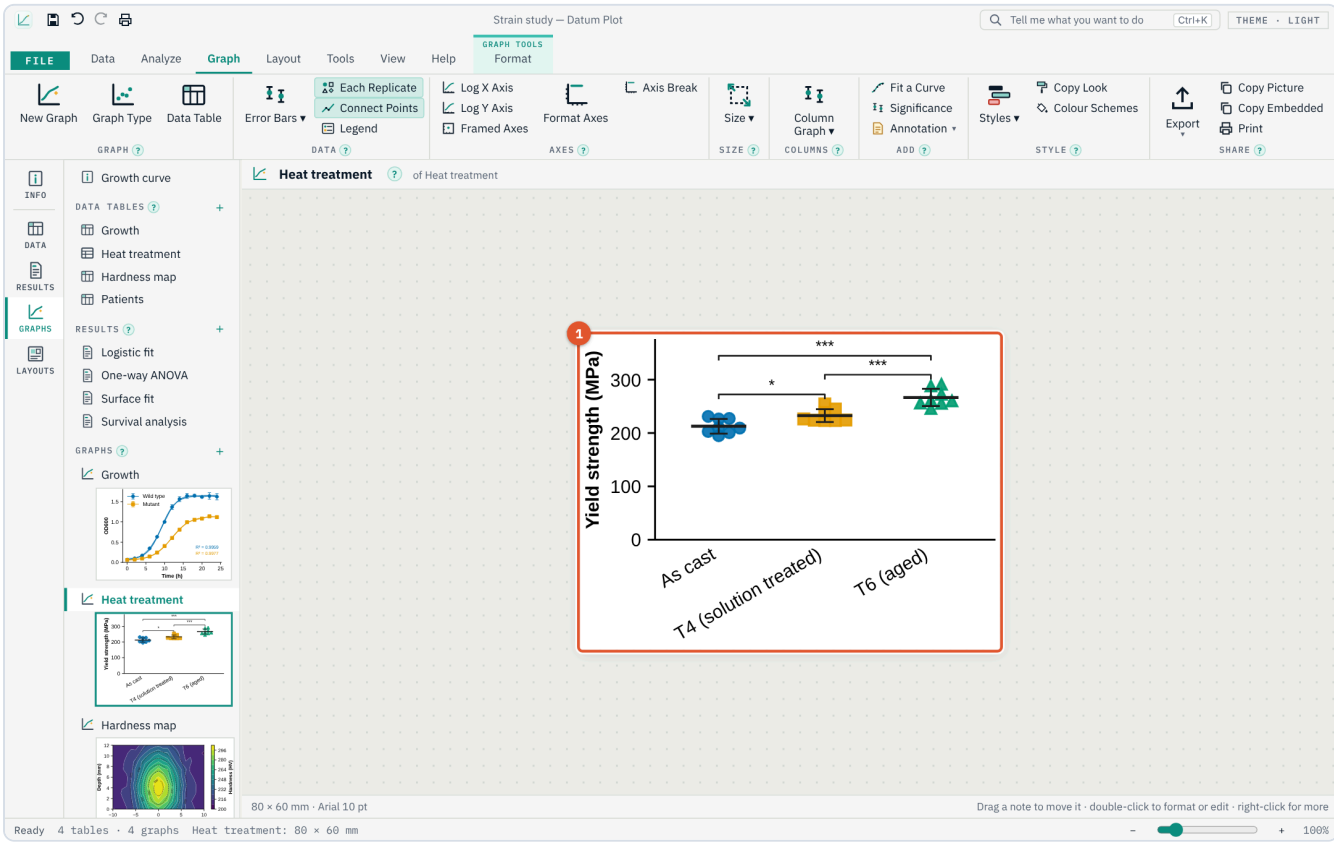
1. **Schemes.** Click one: the graph changes at once, because the window sits beside the graph. Schemes labelled **colour-blind safe** stay easy to tell apart for people with any of the three common kinds of colour-blindness; the default, Okabe-Ito, is one of them.
2. **How colour-blind readers see it.** The three columns show each scheme as people with red-, green- and blue-blindness see it.

Your own colours go in the last row (#hex, separated by commas). In **Format Graph › Data Sets**, each colour is a small square showing the colour as drawn. Click it for the scheme's colours, a colour of your own, the **Strength** of a fill, or **Automatic** (back to the scheme); a square marked A is automatic. On bar, box and violin graphs the bars' fill and border come first, then the symbols' shape, size, fill and border, each set on its own. **Reset all colours** puts every colour back to the scheme. Any window can be moved by dragging its title bar.

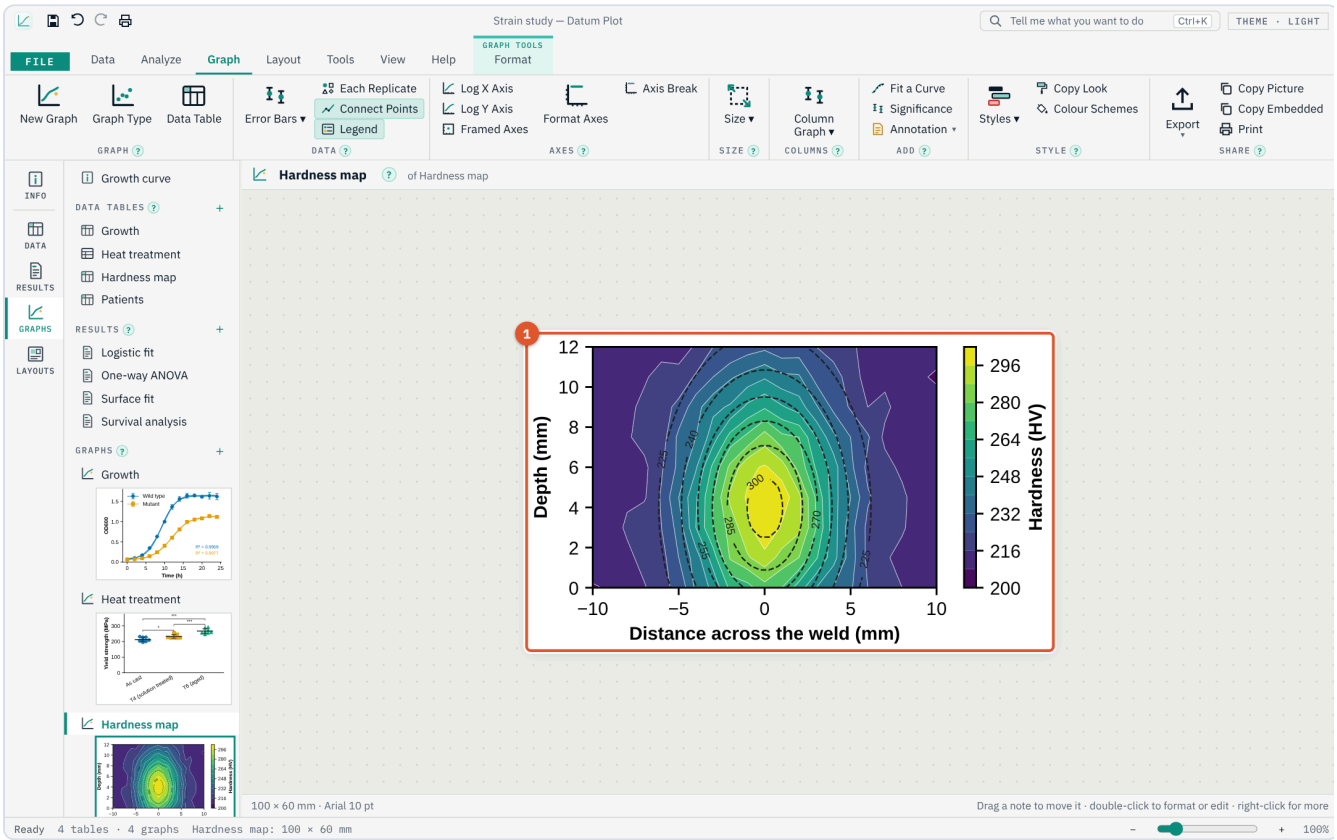
Working on the graph

- Hover a point to see its values.
- Drag a note to move it; double-click it to edit.
- Right-click a point to exclude it.
- **Zoom to a Box**, **Pan** and **Statistics of a Region** are in the graph's right-click menu; `Ctrl` and the mouse wheel zoom the view.

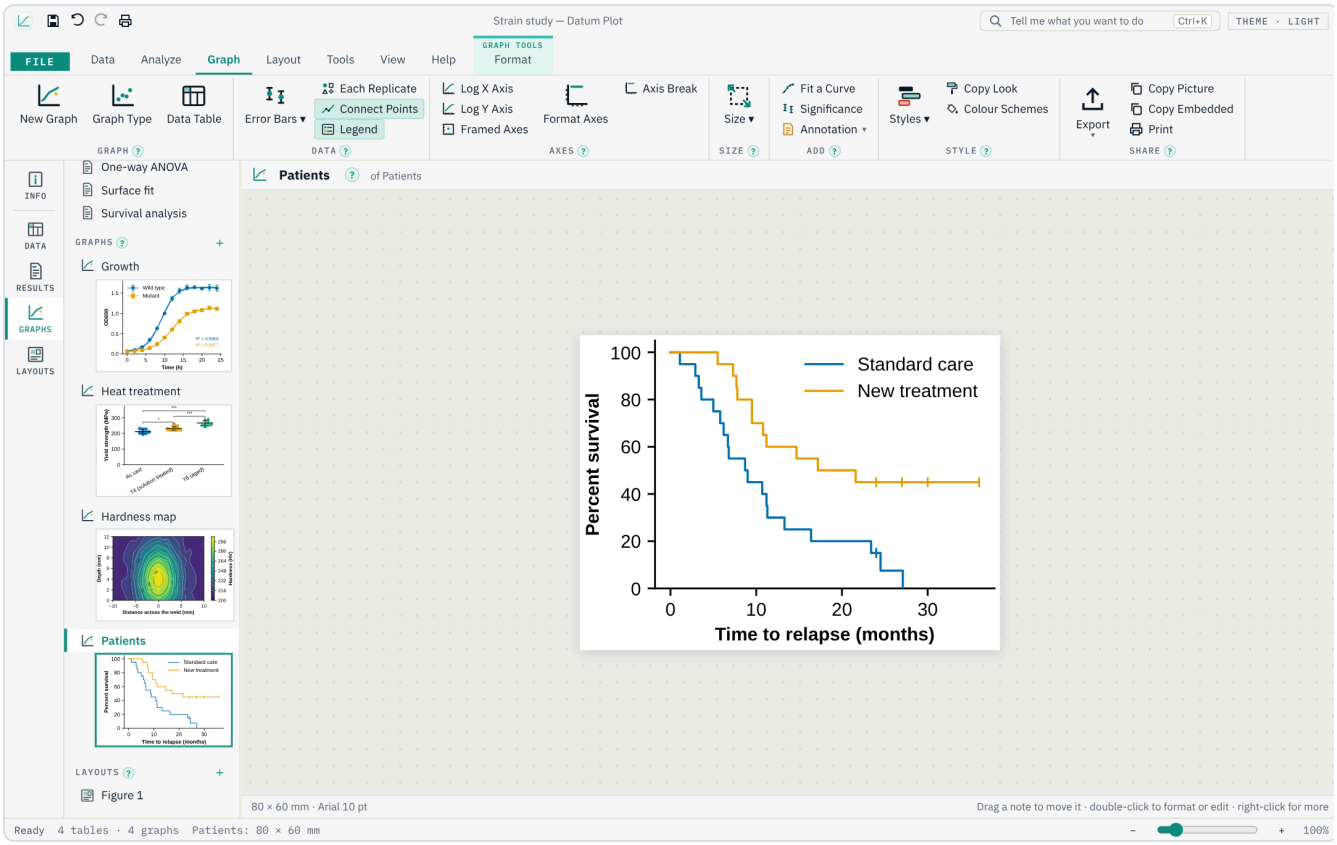
Graphs for every kind of data



Compare groups: each value, mean and SD, with significance stars from the ANOVA.



A matrix table as a contour map with a surface fit.



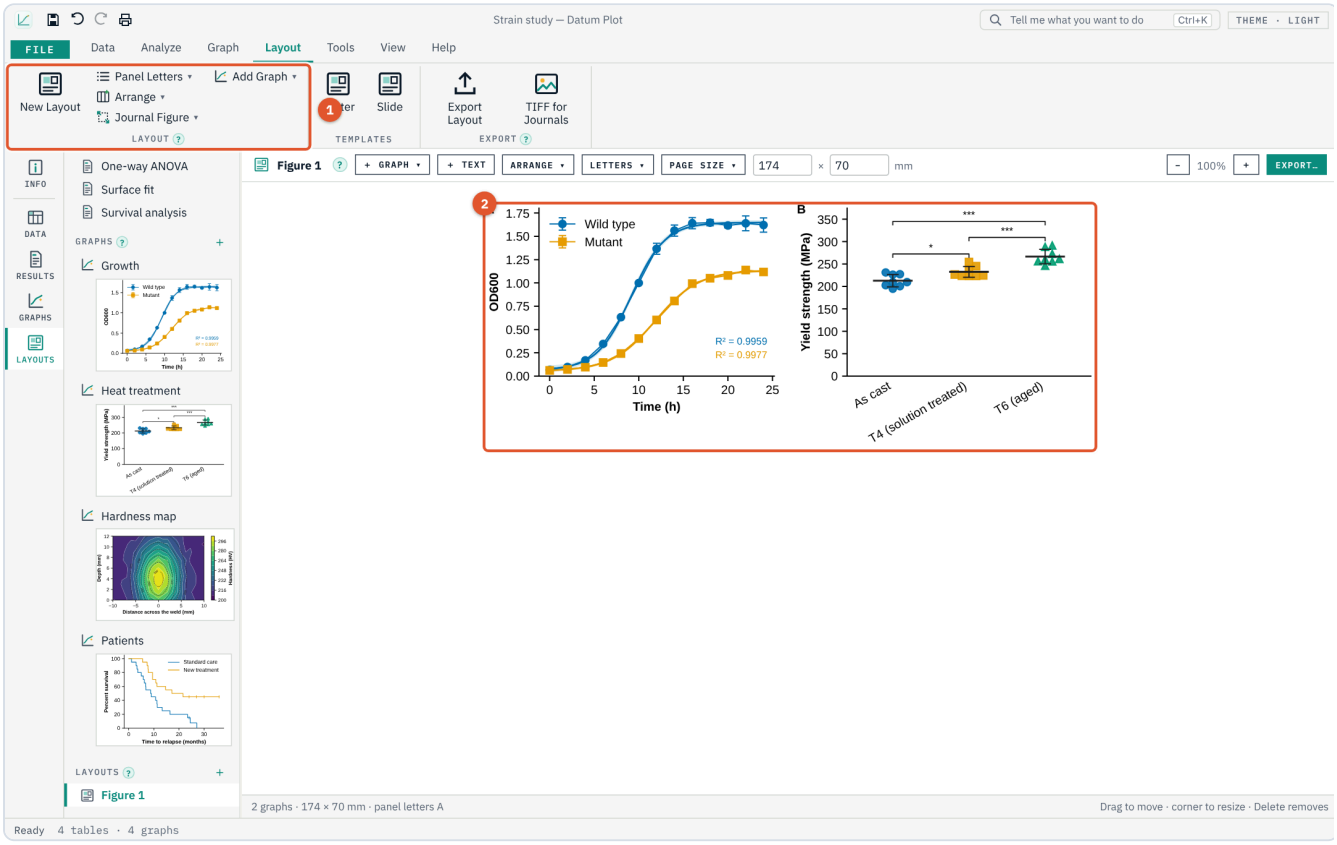
A survival study: Kaplan-Meier curves with censored subjects marked.

XY graphs can also be polar graphs (**Format Graph › Graph › Polar**) or waterfalls, with each curve lifted above the last (**Graph › Graph Style › Waterfall**). Tables of many variables can be ternary graphs (the first three variables as shares of 100 %) bubble graphs (X, Y, bubble size and colour), 3D scatter graphs (X, Y, Z and colour) or vector fields (arrows at X, Y with their two components); grouped tables can be radar graphs (the rows are the spokes). Matrix tables can be drawn as contour maps, heat maps, 3D surfaces or wireframes.

Layouts and export

Page layouts

A layout places several graphs and text on one page: a multi-panel figure for a journal, a poster or a slide.

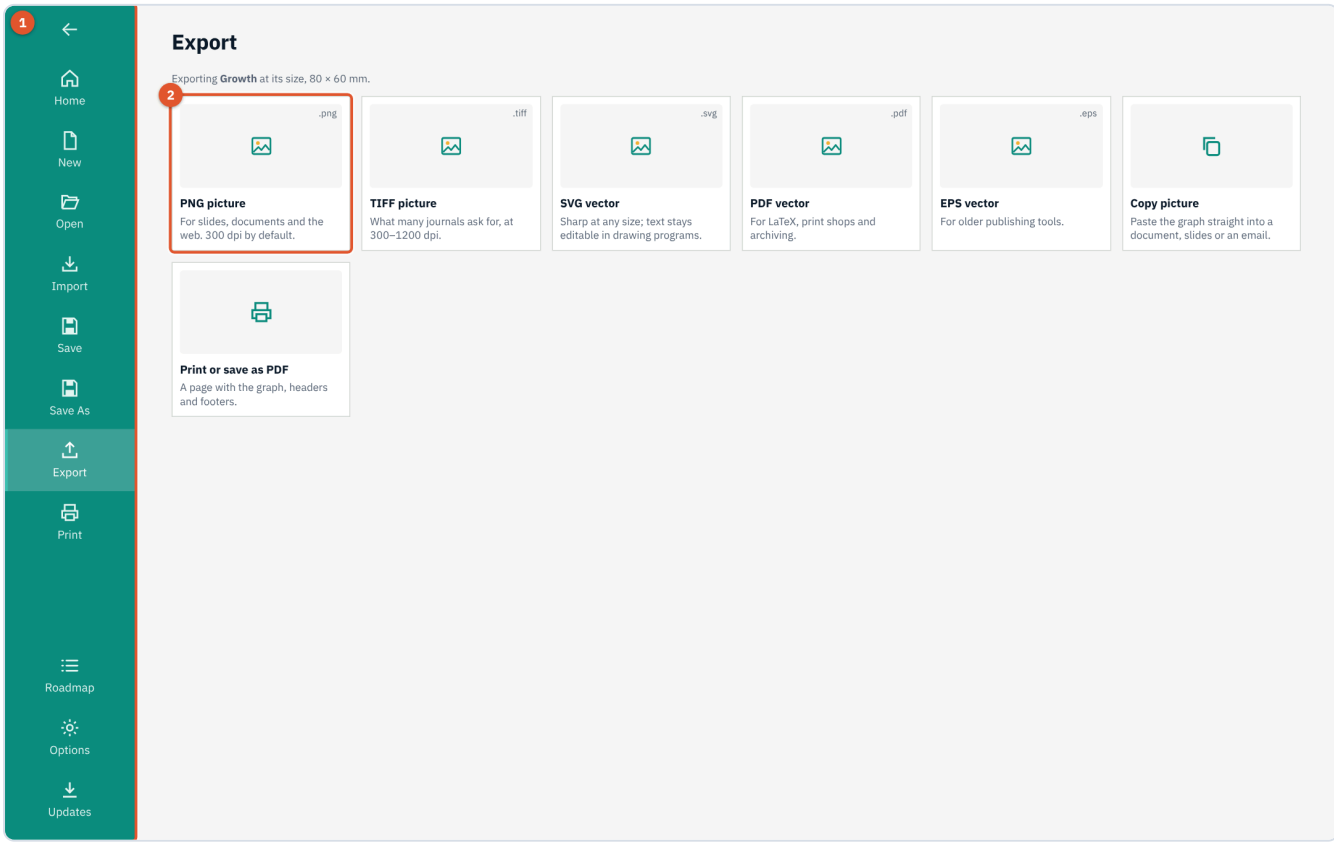


A two-panel figure (Layout tab).

1. **Layout. New Layout**, **Panel Letters** (A, a, (a)), **Arrange** (line up and size panels), **Journal Figure** (a single or double column of a journal) and **Add Graph**.
2. **The page.** Drag a panel to move it; drag a corner to resize it. Right-click a panel for more choices.

The **Templates** group starts a **Poster** or a **Slide**. The **Export** group saves the page.

Export



File › Export: the formats.

1. **File › Export** exports the open graph.
2. **Formats. PNG** for slides and the web; **TIFF** (300–1200 dpi, CMYK if asked) for journals; **SVG**, **PDF** and **EPS** vectors that stay sharp at any size. **Copy picture** puts the graph on the clipboard.

Other ways to share, under **Tools**:

- **Export All Graphs** at once.
- **Workbook**: every table and result as sheets of a .xlsx or .ods file.
- **Slides**: every graph on its own slide, and a slide per result (.pptx or .odp).
- **Document**: the project's details, every graph with a caption, every result and the methods text (.docx or .odt).

Choose the file type in the Save dialog. The .ods, .odt and .odp files are OpenDocument, the open standard.

Graphs you can edit again from a document

Graph › Copy Embedded (or right-click a graph) copies the graph with its data. Paste it into a document, slides or a workbook on Windows: it looks like the picture, and **double-clicking it there opens it in Datum Plot**. Change the data, colours or axes, then click **Update** in the bar at the top (or close the window and choose **Update and Close**): the graph in the document changes. The data travels inside the document, so a colleague with Datum Plot can open it too.

- **Copy Picture** stays the simple choice: a picture that is not linked to Datum Plot.
- On a Mac, or in a browser, the embedded graph shows as its picture and cannot be opened.
- The first graph of the window is the one the document shows.
- In a word processor, paste with **Paste Special › Datum Plot Graph Object** (a plain paste may give a picture).
- If double-clicking does nothing, **Help › Check Embedded Graphs** tests it and says what went wrong.

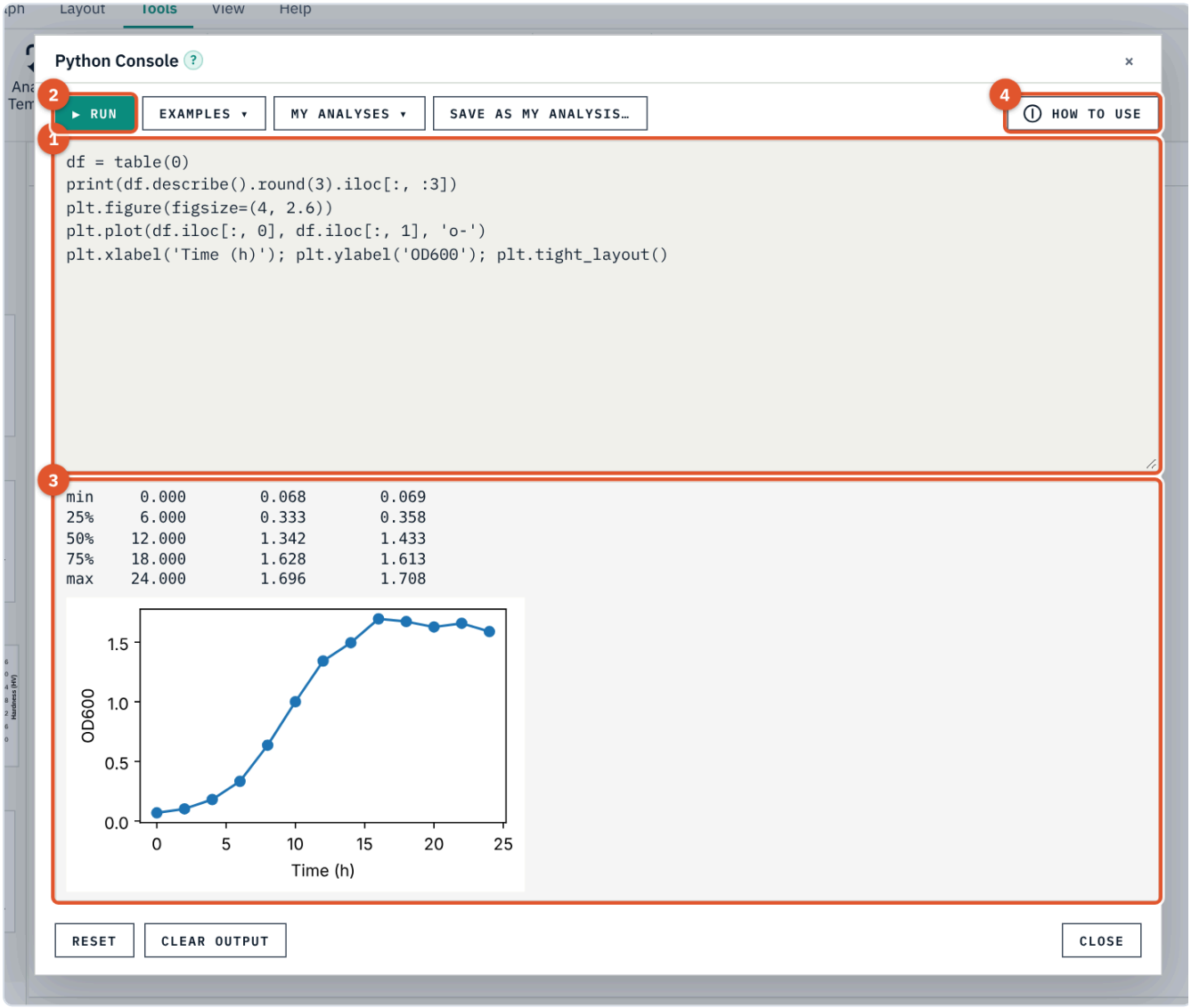
Print

File › Print prints a graph with a header and footer, or saves it as a PDF.

The Python console and automation

Python console

The Python console runs your own Python code on the project's data.



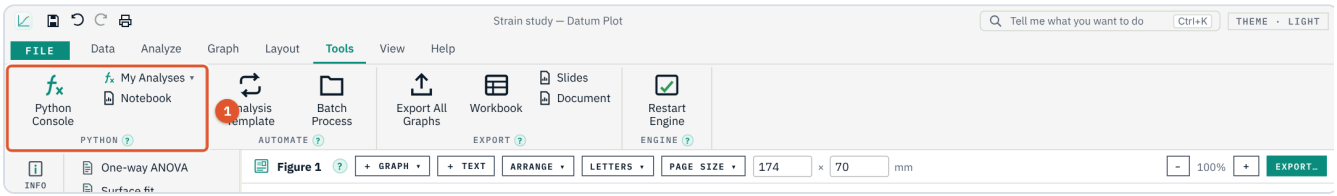
Tools › Python Console.

1. **Code.** Write Python here. Your tables are ready as pandas DataFrames: `tables` lists them, `table(0)` is the first, `tables['Name']` one by its name.
2. **Run** (or `Ctrl+Enter`).
3. **Output.** What you print, the value of the last line, and any figure you draw with `plt`.
4. **How to use.** A short guide inside the console; **Examples** fills in working code to start from.

Ready to use: `np` (numpy), `pd` (pandas), `stats` (scipy.stats), `scipy` and `plt` (matplotlib). `new_table('Name', data)` adds a table to the project; `result('Analysis name')` gives an analysis's numbers. Variables stay between runs until you press **Reset**.

> **Note.** The console runs code you write, with your rights, like any Python script. Projects never contain code: opening a project from someone else never runs anything.

Automation



The Tools tab.

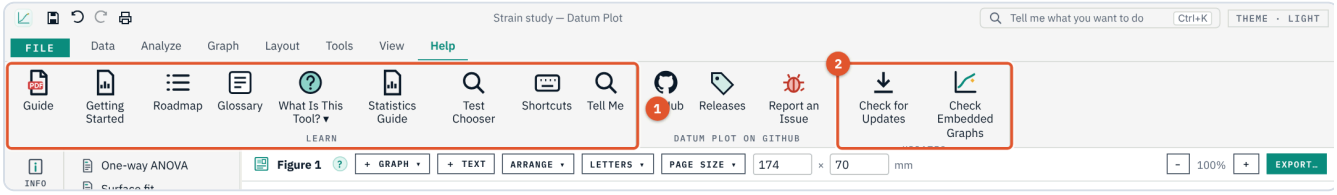
1. **Python. Python Console**, **My Analyses** (code you saved with **Save As My Analysis**) and **Notebook** (the project as a Jupyter notebook).

- **Analysis Template** repeats one table's analyses (with their options) and its graph's look on other tables.
- **Batch Process** takes a template table and many files: each file becomes a table with those analyses and a graph, and the answers are listed together.

Files

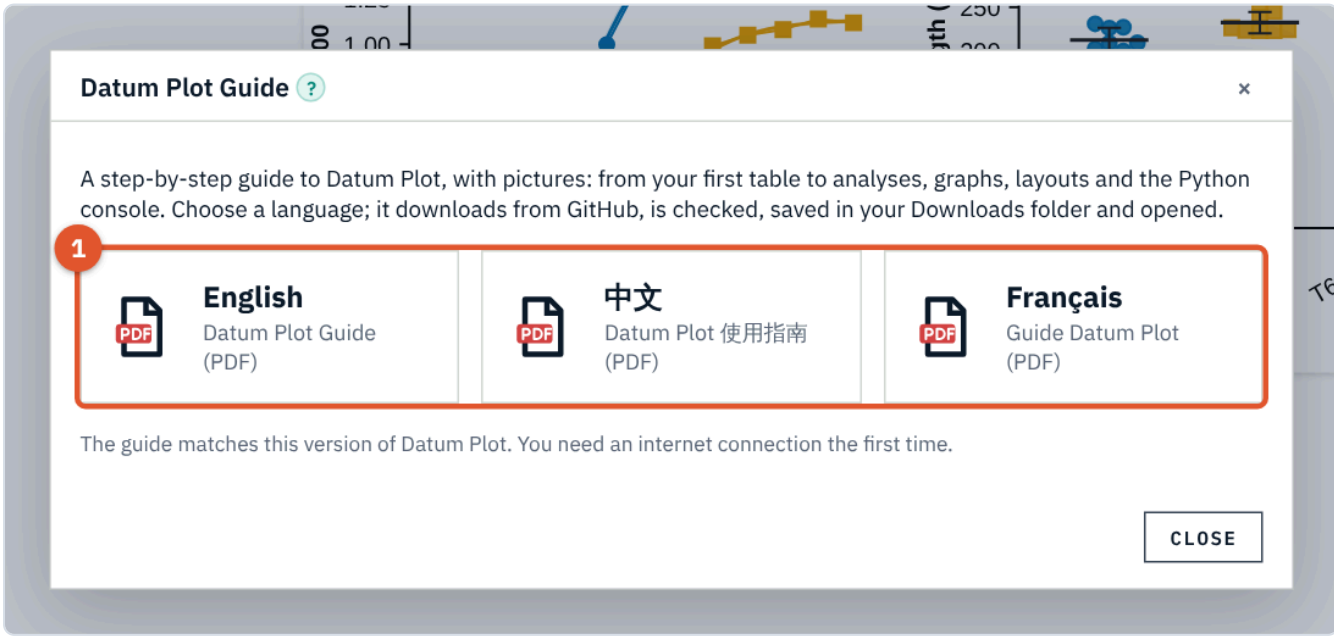
- **Save** (`Ctrl+S`) keeps the whole project in one `.dplot` file: tables, analyses, graphs and layouts. Saving is atomic: a crash or a full disk never leaves a half-written file.
- **Open** (`Ctrl+O`) opens a `.dplot` file; you can also double-click it in Windows.
- Datum Plot saves a recovery copy every few minutes (**File › Options**). If it closes unexpectedly, it offers to recover your work at the next start.
- **Undo** and **Redo** (`Ctrl+Z`, `Ctrl+Y`) go back through every change.

Help in the app



The Help tab.

1. **Learn. Guide** (this guide), **Getting Started**, **Roadmap** (what Datum Plot will do, and when), **Glossary**, **What Is This Tool?**, **Statistics Guide**, **Test Chooser**, **Shortcuts** and **Tell Me**.
2. **Updates. Check for Updates**.



Help › Guide: choose a language.

1. **Languages.** English, 中文 or Français. The guide downloads from GitHub, is checked against its checksum, is saved in your Downloads folder and opens.

Which test when

Your question	Your data	Analysis
Do two groups differ?	two columns of values	t-test (Mann-Whitney if not normal)
Did the same items change?	before and after, paired by row	paired t-test (Wilcoxon if not normal)
Do three or more groups differ?	columns of values	one-way ANOVA with multiple comparisons (Kruskal-Wallis if not normal)
Do two factors matter?	grouped table	two-way ANOVA (repeated measures if the same subjects are measured again)
Are proportions different?	counts	chi-square or Fisher's exact test
How are X and Y related?	XY	linear regression or curve fit
Do two variables move together?	two columns	correlation
Which of many genes or proteins changed?	grouped table, one row each, two data sets	multiple t-tests and a volcano plot
Do two methods agree?	paired values	Bland-Altman
Do two raters agree?	contingency table of paired ratings	Agreement (Cohen's kappa)
Do groups differ once a covariate is allowed for?	XY: covariate in X, one data set per group	ANCOVA
How well does a test detect a condition?	values of controls and cases	ROC curve
How long until an event?	times with events and censored subjects	Kaplan-Meier and log-rank

> **Tip. Analyze › Test Chooser** asks these questions for you.

Glossary

Term	Meaning
Data set	One series of values in a table (a column, or a group of replicate columns).
Replicate	A repeated measurement of the same thing.
SD	Standard deviation: how spread out the values are.
SEM	Standard error of the mean: how precisely the mean is known (SD/$\sqrt{n}$).
95% CI	95% confidence interval: the range that contains the true value with 95% confidence.
P value	How likely a difference this large would be if there were no real difference.
R^2	The share of the variation that a fitted line or curve explains.
Censored	A subject whose event had not happened when it was last seen.
Residual	The distance of a point from the fitted curve.

Shortcuts and troubleshooting

Shortcuts

Keys	Action
`Ctrl+S` / `Ctrl+Shift+S`	Save / Save As
`Ctrl+O` / `Ctrl+N`	Open / New
`Ctrl+Z` / `Ctrl+Y`	Undo / Redo
`Ctrl+C` / `Ctrl+V`	Copy / paste (works with spreadsheets)
`Ctrl+E`	Exclude or include values
`Ctrl+D`	Fill down
`Ctrl+C` on a graph	Copy the graph as a picture
`Ctrl+P`	Print
`Ctrl+K`	Tell me what you want to do
`F1`	All shortcuts

Troubleshooting

- **The status bar says the graph engine is not available.** Click **Tools › Restart Engine**. If it still fails, reinstall Datum Plot from the Releases page; your projects are not affected.
- **An analysis says it cannot be worked out.** The message says why (too few values, identical values, a missing X). Fix the data or choose other data sets with **Change...**.
- **A graph shows no points on a log axis.** Values of zero or below cannot be drawn on a log axis; **Data › Check Data** finds them.
- **Something else went wrong.** That step is refused and your project stays as it was; the details are in `crash.log` in the app's data folder. Please report it with **Help › Report an Issue**.